Berkay Barlas

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Summary

Software Engineer specializing in mission-critical consumer software systems, with expertise in browser platforms, regulatory compliance, and AI-driven features. As a core engineer on Apple's Safari iOS team, I support a browser serving one billion+ users, with key responsibility for DMA compliance initiatives and AI integration. My contributions include patent-pending technologies and peer-evaluation service as a jury member for Apple's Swift Student Challenge (2025).

Education

Georgia Institute of Technology (GaTech) — Atlanta, US

M.Sc Computer Science — Aug 2021 - Jun 2023

GPA 4.0 / 4.0 | Selected for and completed advanced coursework in machine learning systems, networks, and large-scale distributed computing, directly relevant to browser and AI platform development.

Koc University — Istanbul, Turkey

B.S. Computer Eng. & Electronics Eng. Double Major — Oct. 2015 - Jun. 2021

- **GPA 3.8 / 4.0** | Received 100% scholarship which is given to top 300(%0.01) among 2 million students.
- **Related Coursework:** Deep Learning, High Performance Parallel Programming, Modern Cryptography

Technical University of Munich (TUM) — Munich, Germany

Exchange student with Erasmus+ Full Grant — Apr. 2020 - Sept. 2020

- **Related Coursework:** Advanced Deep Learning, Machine Learning for Computer Vision, Blockchain-based Systems

Nanyang Technological University (NTU) — Singapore

Exchange student with Full-Scholarship — Aug. 2019 - Dec. 2019

- **Related Coursework:** Artificial Intelligence and Data Mining

Full-Time Experience

Apple — Cupertino, CA

Software Engineer — 2+ Year | July 2023 - Current

- Core engineer on the Safari iOS browser, a system application serving 1B+ users worldwide.
- Primary implementation contributor for EU Digital Markets Act (DMA) compliance, delivering mandatory regulatory changes across Safari and iOS system layers.
- Designed AI-driven browser features, balancing performance, privacy, and large-scale deployment requirements.
- Worked within a small, highly specialized team, each engineer owns production-critical components affecting global releases.
- Work directly impacts regulatory compliance, platform stability, and user experience for Safari deployments across the United States and European Union.

Vagon — Remote

Full Stack Software Developer — 1+ Years | Mar 2020 - Aug 2021

- Maintained and optimized full-stack applications, leveraging React, Ruby on Rails, Postgre SQL, AWS, Docker.
- Enhanced live streaming service performance **30%** using WebAssembly, WebSocket, TCP/IP, UDP protocols and browser APIs.
- Reduced monthly fraud cases 80% by implementing improved payment systems with fraud detection, prevention methods.

Mozilla / Firefox Devtools — Remote

Open Source Contributor — 3 Months | Jan 2020 - Mar 2020

- Identified and fixed bugs on Github and Bugzilla(Phabricator) under **Firefox** projects, mainly on **Firefox Devtools** and Gecko.
- Reworked Firefox reader mode design with Sketch and integrated reader-specific buttons to work with browser settings.

Zeplin — Istanbul, Turkey

Software Engineer — 6 Months | June. 2019 - Oct. 2019

- Early engineer at Zeplin, a Y Combinator–backed design-to-development platform used by global product teams.
- Owned core front-end and Electron architecture used across production applications.
- Led performance refactors, reducing code redundancy and improving page load performance by 30%+.

Trendyol.com — Istanbul, Turkey

Software Developer Part-Time — 6 Months | Jul. 2018 - Dec. 2018

- Developed web services for seller interface of e-commerce application using Java Spring Boot, REST API, RabbitMQ.
- Implemented new microservice features to Seller Portal that are used for adding and tracking products in marketplace.

Internship Experience

Google — San Francisco, CA

Software Engineering Intern — 4 Months | Aug 2022 - Nov 2022

- Developed a tool to track and identify latency changes in Google Cloud products and diagnose potential bottlenecks.
- Implemented a TypeScript dashboard and Go back-end to aggregate billions of logs using SQL queries and visualize data.
- Created documents to prepare team for migrating to more modern internal frameworks and organized workshops.

Apple — Cupertino, CA

Software Engineering Intern — 4 Months | May 2022 - Aug 2022

- Designed and implemented key features in an in-house recommendation system, improving predictive accuracy for incomplete search inputs by 4%.
- Collaborated with Safari and Webkit team members to enhance the functionality and reliability of the search system.
- Resolved critical Safari bugs using UIKit, AppKit, Objective-C, contributing to a smoother user experience and reduced crashes.

Research & Teaching Experience

Graduate Teaching Assistant, Georgia Institute of Technology

OMCS: Advanced Internet Computing Systems — Spring'22

- Created assignments and grading rubrics; regularly answered student questions on course website.
- Graded weekly assignments and led bi-weekly Q&A sessions, improving student understanding of web systems.

Network and Distributed Systems Laboratory, Koc University

Advisor: Prof. Dr. Öznur Özkasap — Jan. 2019 - Jun. 2019

- Contributed to the Skip Graph, blockchain and distributed system simulator, by developing node search algorithms.

Stanford University CS Bridge Program

Section Leader | Lecturer: Prof. Nick McKeown — Summer'17, Summer'18

- Guided 50+ high school students each year in their introduction to programming and led laboratory sessions.

Patents

Patent Application (Filed, Pending) 2025 — Co-Inventor. System and method for intelligent interface organization using on-device AI in large-scale consumer software platforms (Apple Inc.).

Program Committees

2025 **Judge / Jury Member - Apple Swift Student Challenge (2025)**, Selected by Apple to evaluate global student submissions based on technical quality, creativity, and execution.

Honors & Awards

2019 **Pixery Hackathon'19 Best Project Award**, Ranked 1st among 12 finalist teams(top <0.1% of applicants), selected from thousands of nationwide applicants in Turkey.

2019 **Ciceksepeti Hackathon'19 Best Project Award**, Ranked 1st among 30 finalist teams, selected from thousands of applicants nationwide.

2018 **Zeplin Hackathon Special Jury Award**, Jury-selected award recognizing exceptional technical execution.

2015 **Ranked 273th (%0.01) among more than 2 million Students**, Turkish University Attainment Exam

2011 **Ranked 1st (%0.0001) among more than 1 million Students**, Turkish High School Attainment Exam

Skills

Programming Languages	C / C++, Objective-C, Swift, TypeScript, Kotlin, WebAssembly, HTML, CSS
Web & Application Development	UIKit, SwiftUI, AppKit,React, Node.js, GO
Development Tools	Xcode, Instruments, Firefox Profiler, Safari, Firefox, and Chrome developer tools, Sketch, Keynote

Leadership & Volunteer Experience

Stanford University Hasso Plattner Institute of Design (d.school)

University Innovation Fellow — Oct. 2019 - 2021

- Completed a 7-week-long design thinking and innovation course, brainstormed and launched new ideas.
- Planned and piloted an orientation program with faculty to enhance new students' understanding of university resources.

Global Learning and Leadership Program

Cofounder of Koc University Makerspace Lab — Sept. 2016 - 2021

- Secured funding from the engineering faculty to design and equip the Makerspace, after a one-year pivot project.
- Led a team of 15 members, organizing workshops on computer science, and creative skills for over 200 participants.
- By Spring 2019, the Makerspace supported 40% of all senior engineering projects as a student-led innovation hub.